Lesson 38: Signaling, Perfect Bayesian Equilibrium, and Potential Games

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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7 Perfect Bayesian Equilibrium

7.1 Motivation

In dynamic games of incomplete information, the BNE concept is too weak: it does not constrain behavior at off-equilibrium information sets. We need a refinement that combines sequential rationality with belief consistency.

Definition 7.1 (System of Beliefs)**.** In an extensive-form game with incomplete information, a *system of beliefs* μ assigns to each information set h a probability distribution $\mu(\cdot|h)$ over the nodes in that information set.

Definition 7.2 (Perfect Bayesian Equilibrium)**.** A *perfect Bayesian equilibrium* (PBE) is a pair (σ, μ) of a behavioral strategy profile σ and a belief system μ such that:

(a) **Sequential rationality:** At every information set h of every player i, the strategy σ_i is optimal given beliefs $\mu(\cdot|h)$ and the continuation strategies σ_{-i}.

(b) **Belief consistency:** At every information set h that is reached with positive probability under σ, the beliefs $\mu(\cdot|h)$ are derived from σ using Bayes' rule:

$$\mu(x|h) = \frac{\mathbb{P}(x \text{ is reached}|\sigma)}{\sum_{x' \in h} \mathbb{P}(x' \text{ is reached}|\sigma)} \quad \text{for all } x \in h.$$

(c) At information sets reached with zero probability (off the equilibrium path), beliefs are unrestricted (any distribution over nodes in the information set is allowed).

7.2 Signaling Games

A particularly important class of dynamic Bayesian games is *signaling games*.

Definition 7.3 (Signaling Game)**.** A *signaling game* has two players:

- **Sender (S):** Has private type $\theta \in \Theta$, drawn from prior $p(\theta)$. Observes θ, then sends a message $m \in M$.
- **Receiver (R):** Observes m (but not θ), then takes action $a \in A$.

Payoffs $u_S(\theta, m, a)$ and $u_R(\theta, m, a)$ depend on all three: type, message, and action.

Definition 7.4 (Equilibrium Types in Signaling Games)**.** (a) A PBE is *separating* if different types send different messages: $\sigma_S(\theta) \neq \sigma_S(\theta')$ for $\theta \neq \theta'$.

(b) A PBE is *pooling* if all types send the same message: $\sigma_S(\theta) = m^*$ for all θ.

(c) A PBE is *semi-separating* (or partially pooling) if some types pool and others separate.

Example 7.1 (Spence's Job Market Signaling)**.** A worker has ability $\theta \in \{L, H\}$ with $\mathbb{P}(\theta = H) = \pi$. The worker chooses education level $e \geq 0$ (the "signal"). An employer observes e and offers wage $w(e)$. Payoffs:

$$u_S(\theta, e, w) = w - c(\theta, e), \quad \text{with } c(H, e) < c(L, e) \text{ for } e > 0,$$
$$u_R(\theta, e, w) = \theta - w,$$

where $c(\theta, e)$ is the cost of education (cheaper for high-ability workers: *single-crossing property*).

Separating equilibrium. In a separating PBE, $e_H \neq e_L$ and the receiver's beliefs at e_H are $\mu(H|e_H) = 1$, at e_L are $\mu(H|e_L) = 0$ (by Bayes' rule). The receiver offers $w(e_H) = H$ and $w(e_L) = L$.

For this to be an equilibrium, each type must prefer its own signal:

$$H - c(H, e_H) \geq L - c(H, e_L) \quad \text{(type } H \text{ prefers } e_H\text{)},$$
$$L - c(L, e_L) \geq H - c(L, e_H) \quad \text{(type } L \text{ prefers } e_L\text{)}.$$

The single-crossing property ensures that such e_H, e_L exist (typically $e_L = 0$ and e_H above a threshold).

Pooling equilibrium. Both types choose $e^* = 0$. The receiver's belief at e^* is the prior: $\mu(H|e^*) = \pi$, so $w(e^*) = \pi H + (1 - \pi)L$. Off-equilibrium beliefs at $e \neq e^*$ can be chosen to deter deviation, e.g., $\mu(H|e) = 0$ for all $e \neq e^*$, giving $w(e) = L$. Then no type wants to deviate if $\pi H + (1 - \pi)L - c(\theta, 0) \geq L - c(\theta, e)$ for all $e > 0$, which holds when the pooling wage is sufficiently attractive.

Proposition 7.1 (PBE Existence in Finite Signaling Games)**.** *Every finite signaling game (finite Θ, M, A) has a perfect Bayesian equilibrium.*

Proof. A finite signaling game is a finite extensive-form game. The sequential equilibrium existence theorem of Kreps and Wilson (1982) guarantees existence of a sequential equilibrium, which is a refinement of PBE. In particular, every sequential equilibrium is a PBE, so PBE exists.

Alternatively, one can construct a PBE directly: consider all possible sender strategies $\sigma_S : \Theta \to \Delta(M)$. For each such strategy, compute the receiver's beliefs via Bayes' rule at information sets reached with positive probability, and assign arbitrary beliefs at unreached information sets. The receiver's optimal response defines a best-response map. The sender's optimal message choice, given the receiver's response function, defines another best-response map. By Kakutani's fixed point theorem (or Nash's theorem applied to the agent normal form), a fixed point—hence a PBE—exists. □

8 Potential Games

8.1 Definitions

Potential games form a special class of games where all players' incentives can be captured by a single "potential function." This structure has powerful implications for existence of pure equilibria and convergence of learning dynamics.

Definition 8.1 (Exact Potential Game)**.** A finite game $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$ is an *exact potential game* if there exists a function $\Phi : A \to \mathbb{R}$ (the *potential function*) such that for every player i, every $a_{-i} \in A_{-i}$, and every $a_i, a_i' \in A_i$:

$$u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i}).$$

Definition 8.2 (Ordinal Potential Game)**.** A finite game G is an *ordinal potential game* if there exists $\Phi : A \to \mathbb{R}$ such that for every player i, every a_{-i}, and every a_i, a_i':

$$u_i(a_i, a_{-i}) > u_i(a_i', a_{-i}) \iff \Phi(a_i, a_{-i}) > \Phi(a_i', a_{-i}).$$

Remark 8.1. Every exact potential game is an ordinal potential game. The converse is false: ordinal potentials only preserve the *sign* of payoff differences, not their magnitude.

8.2 Existence of Pure Nash Equilibria

Theorem 8.1 (Potential Games Have Pure NE)**.** *Every finite exact potential game (and every finite ordinal potential game) has at least one pure Nash equilibrium.*

Proof. Let $\Phi : A \to \mathbb{R}$ be the potential function. Since A is finite, Φ attains its maximum at some $a^* \in A$:

$$a^* \in \operatorname{argmax}_{a \in A} \Phi(a).$$

We claim a^* is a pure Nash equilibrium.

For any player i and any deviation $a_i' \in A_i$:

$$\Phi(a_i^*, a_{-i}^*) \geq \Phi(a_i', a_{-i}^*)$$

by the maximality of a^*. By the potential property:

$$u_i(a_i^*, a_{-i}^*) - u_i(a_i', a_{-i}^*) = \Phi(a_i^*, a_{-i}^*) - \Phi(a_i', a_{-i}^*) \geq 0.$$

Thus a_i^* is a best response to a_{-i}^* for every player i, so a^* is a NE.

For ordinal potential games, the same argument works: $\Phi(a_i^*, a_{-i}^*) \geq \Phi(a_i', a_{-i}^*)$ implies $u_i(a_i^*, a_{-i}^*) \geq u_i(a_i', a_{-i}^*)$ by the ordinal potential property (since if the strict inequality failed, i.e., $u_i(a_i', a_{-i}^*) > u_i(a_i^*, a_{-i}^*)$, then $\Phi(a_i', a_{-i}^*) > \Phi(a_i^*, a_{-i}^*)$, contradicting maximality). □

8.3 Convergence of Best-Response Dynamics

Definition 8.3 (Best-Response Dynamics)**.** In *(asynchronous) best-response dynamics*, players take turns updating their actions. At each step, one player i is selected (deterministically or randomly) and switches to a best response to the current actions of all other players:

$$a_i^{t+1} \in \operatorname{argmax}_{a_i \in A_i} u_i(a_i, a_{-i}^t), \qquad a_j^{t+1} = a_j^t \text{ for } j \neq i.$$

Theorem 8.2 (Finite Improvement Property)**.** *In every finite exact potential game, best-response dynamics converge to a pure Nash equilibrium in a finite number of steps.*

Proof. We show that Φ strictly increases at each step where a player's action changes, hence the dynamics must terminate (since A is finite and Φ cannot increase indefinitely).

Suppose at step t, player i switches from a_i^t to $a_i^{t+1} \neq a_i^t$ with $a_i^{t+1} \in \operatorname{argmax}_{a_i} u_i(a_i, a_{-i}^t)$. Since a_i^{t+1} is a strictly improving move (otherwise the player would not switch), we have:

$$u_i(a_i^{t+1}, a_{-i}^t) > u_i(a_i^t, a_{-i}^t).$$

By the exact potential property:

$$\Phi(a_i^{t+1}, a_{-i}^t) - \Phi(a_i^t, a_{-i}^t) = u_i(a_i^{t+1}, a_{-i}^t) - u_i(a_i^t, a_{-i}^t) > 0.$$

Thus $\Phi(a^{t+1}) > \Phi(a^t)$ whenever an action changes. Since A is finite, the sequence $\Phi(a^1), \Phi(a^2), \ldots$ is strictly increasing and takes values in the finite set $\{\Phi(a) : a \in A\}$, so it must terminate. Termination occurs when no player wants to change, i.e., at a pure Nash equilibrium.

Remark on step count: The number of steps is bounded by $|A| = \prod_i |A_i|$ (since each action profile can appear at most once in the improvement path), but this worst-case bound is typically very loose. □

Corollary 8.3. *The finite improvement property also holds for* better-response dynamics *(where player i switches to any action that improves her payoff, not necessarily the best response), since Φ still strictly increases.*

Proof. If $u_i(a_i', a_{-i}^t) > u_i(a_i^t, a_{-i}^t)$, then $\Phi(a_i', a_{-i}^t) > \Phi(a_i^t, a_{-i}^t)$ by the potential property. The same termination argument applies. □

8.4 Congestion Games

The most important class of potential games in applications is congestion games.

Definition 8.4 (Congestion Game)**.** A *congestion game* is defined by $(N, E, (A_i)_{i\in N}, (d_e)_{e\in E})$ where:

- $N = [n]$ is the player set.
- E is a finite set of *resources* (or *facilities*).
- $A_i \subseteq 2^E$ is a collection of subsets of resources available to player i. Each action $a_i \in A_i$ is a set of resources that player i uses.
- $d_e : \{1, \dots, n\} \to \mathbb{R}$ is the *delay function* (or cost function) for resource e, depending on the number of users.

The payoff (negative cost) to player i under action profile $a = (a_1, \dots, a_n)$ is:

$$u_i(a) = -\sum_{e\in a_i} d_e\big(n_e(a)\big),$$

where $n_e(a) = |\{j \in N : e \in a_j\}|$ is the congestion (number of users) of resource e.

Theorem 8.4 (Congestion Games are Exact Potential Games (Rosenthal))**.** *Every congestion game is an exact potential game with potential function*

$$\Phi(a) = -\sum_{e\in E}\sum_{k=1}^{n_e(a)} d_e(k).$$

Proof. We must verify that for any player i, any a_{-i}, and any $a_i, a_i' \in A_i$:

$$u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i}).$$

Consider the left side. Let $n_e = n_e(a_i, a_{-i})$ denote the congestion when player i uses a_i, and $n_e' = n_e(a_i', a_{-i})$ when player i uses a_i'. We partition the resources into three groups:

(i) $e \in a_i \cap a_i'$: used by i under both actions. Then $n_e = n_e'$, so $d_e(n_e)$ cancels in both u_i and Φ differences.

(ii) $e \in a_i \setminus a_i'$: used by i under a_i but not a_i'. Then $n_e = n_e' + 1$ (one more user under a_i). The contribution to $u_i(a_i, a_{-i}) - u_i(a_i', a_{-i})$ is $-d_e(n_e) = -d_e(n_e' + 1)$. The contribution to $\Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i})$ is $-\sum_{k=1}^{n_e'+1} d_e(k) + \sum_{k=1}^{n_e'} d_e(k) = -d_e(n_e' + 1)$. These are equal.

(iii) $e \in a_i' \setminus a_i$: used by i under a_i' but not a_i. Then $n_e' = n_e + 1$. The contribution to $u_i(a_i, a_{-i}) - u_i(a_i', a_{-i})$ is $0 - (-d_e(n_e')) = d_e(n_e + 1)$. The contribution to $\Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i})$ is $-\sum_{k=1}^{n_e} d_e(k) + \sum_{k=1}^{n_e+1} d_e(k) = d_e(n_e + 1)$. These are equal.

Summing over all resources, $u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i})$. □

Corollary 8.5. *Every finite congestion game has a pure Nash equilibrium, and best-response dynamics converge to one in finitely many steps.*

Proof. Immediate from Theorems 8.4, 8.1, and 8.2. □

Example 8.1 (Network Routing Game)**.** Consider a network with source s and destination d, and n users each choosing a path from s to d. Each edge e has delay $d_e(k)$ depending on congestion k. This is a congestion game: resources are edges, each player's action set is the set of s-d paths, and the total delay on a path is $\sum_{e\in\text{path}} d_e(n_e)$. By Corollary 8.5, a pure NE exists and can be found by best-response dynamics.

Example 8.2 (Resource Allocation)**.** Consider n agents each choosing one of m machines to process a job. Machine e has processing time $d_e(k) = k \cdot \ell_e$ (linear congestion) where ℓ_e is the base load. Each agent's cost is the processing time on her chosen machine. This is a congestion game with $A_i = E = \{1, \ldots, m\}$ (each action uses a single resource). The potential function is $\Phi(a) = -\sum_{e=1}^{m} \ell_e \cdot \frac{n_e(n_e+1)}{2}$, and pure NE exist with convergent best-response dynamics.

Potential Games and Independent Learning in MARL

Potential games are among the few game classes where independent learners (agents running separate RL algorithms without coordination) are guaranteed to converge:

- **Best-response dynamics** correspond to each agent independently computing optimal policies given fixed opponent policies. The finite improvement property ensures convergence.
- **Log-linear learning:** If each agent plays action a_i with probability proportional to $\exp(\beta \cdot u_i(a_i, a_{-i}))$ (Boltzmann exploration), the joint process is a Markov chain that converges to the potential maximizer as $\beta \to \infty$.
- **Multi-agent reward shaping:** If a team reward function Φ can serve as a potential for individual reward functions u_i, then shaping individual rewards via Φ makes the game a potential game, guaranteeing convergence.

This is the theoretical basis for *difference rewards* and *Wonderful Life Utility* in cooperative MARL.

9 Connections to Multi-Agent RL

We consolidate the connections between the game-theoretic results of this lesson and modern multi-agent reinforcement learning.

Independent Learners in Repeated Games

When n agents independently run Q-learning or policy gradient in a repeated game, each agent treats the others as part of the environment. The folk theorems (Theorems **??** and **??**) imply that the set of stable outcomes is vast—any feasible, individually rational payoff can be a fixed point. This explains the well-known instability and sensitivity to hyperparameters of independent learners in general-sum games. Convergence guarantees for independent Q-learning exist only in restrictive settings: two-player zero-sum games (minimax-Q), potential games, and certain classes of team games.

Self-Play and Fictitious Play

Self-play—where an agent trains against copies of itself—is a form of fictitious play. In zero-sum games, Robinson's theorem (Theorem **??**) guarantees that the empirical strategy profile converges to NE. Modern algorithms extend this:

- **Neural Fictitious Self-Play (NFSP):** Uses neural networks to approximate best responses and average strategies, converging to approximate NE in large zero-sum games.
- **PSRO (Policy-Space Response Oracles):** Maintains a population of policies and computes a best response to the meta-mixture, directly implementing fictitious play in policy space.

- **Double Oracle:** Iteratively adds best responses to a restricted game, converging to the NE of the full game.

The regret bound of MWU (Theorem **??**) provides the convergence rate: $O(1/\sqrt{T})$ for last-iterate convergence in zero-sum games with optimistic updates.

Opponent Modeling as Bayesian Inference

The Bayesian game framework (Section **??**) provides the foundation for opponent modeling in MARL:

(i) **Type inference:** An agent maintains a belief $p(\theta_{-i}|h^t)$ over opponent types, updated via Bayes' rule as actions are observed.

(ii) **Bayesian policy optimization:** The agent's policy maximizes expected return under the current belief: $\max_{\pi_i} \mathbb{E}_{\theta_{-i}\sim p(\cdot|h^t)}[R_i(\pi_i, \pi_{-i}(\theta_{-i}))]$.

(iii) **Exploration-exploitation:** The agent may take actions to learn about opponents' types (information-gathering exploration), connecting to the theory of Bayesian exploration in MDPs.

Potential-Based Reward Shaping in Multi-Agent Settings

Ng, Harada, and Russell's (1999) potential-based reward shaping theorem states that adding a shaping reward $F(s, s') = \gamma\phi(s') - \phi(s)$ preserves the optimal policy in single-agent MDPs. In multi-agent settings, this connects to potential games:

- If the joint reward function serves as a potential for individual rewards, the multi-agent system is a potential game and independent learning converges.
- The *difference reward* $D_i(a) = R(a) - R(a_{-i}, c_i)$ (where c_i is a default action for agent i) is designed so that $D_i(a_i, a_{-i}) - D_i(a'_i, a_{-i}) = R(a_i, a_{-i}) - R(a'_i, a_{-i})$, making R an exact potential for the game with payoffs D_i.
- This theoretical property ensures that maximizing individual difference rewards aligns with maximizing the global objective, providing a principled approach to cooperative MARL.

10 Exercises

Exercise 10.1 (Finitely Repeated Prisoner's Dilemma)**.** Consider the Prisoner's Dilemma from Example **??**, repeated $T = 100$ times. Show that the unique SPE is mutual defection in every period, regardless of T. Explain why this result is "paradoxical" and how it motivates infinitely repeated games.

Exercise 10.2 (Grim Trigger in a Different Game)**.** Consider the stage game:

	L	C	R
T	$6,6$	$0,8$	$0,0$
M	$8,0$	$4,4$	$0,0$
B	$0,0$	$0,0$	$2,2$

(a) Find all pure Nash equilibria of the stage game.

(b) Compute the minimax values $\underline{v}_1$ and $\underline{v}_2$.

(c) Show that the payoff $(6, 6)$ (from (T, L)) can be sustained as an SPE of the infinitely repeated game using grim trigger with punishment (B, R). Find the minimum discount factor.

(d) Can the payoff $(7, 7)$ be sustained? If so, construct a strategy profile.

Exercise 10.3 (Tit-for-Tat Is Not SPE)**.** Show that in the infinitely repeated Prisoner's Dilemma (Example **??**), the tit-for-tat strategy profile is a Nash equilibrium for $\delta \geq 1/3$ but is *not* a subgame perfect equilibrium. (Hint: Consider the subgame after (D, C).)

Exercise 10.4 (Computing BNE)**.** Consider a two-player Bayesian game where player 1 has type $\theta_1 \in \{H, L\}$ with $\mathbb{P}(\theta_1 = H) = 1/2$, and player 2 has no private information. Actions: $A_1 = A_2 = \{U, D\}$. Payoffs (for each type of player 1):

Type H :

	L	R
U	$2, 1$	$0, 0$
D	$0, 0$	$1, 2$

Type L :

	L	R
U	$0, 0$	$1, 2$
D	$2, 1$	$0, 0$

Find all Bayesian Nash equilibria (pure and mixed).

Exercise 10.5 (Second-Price Auction as Bayesian Game)**.** In a second-price sealed-bid auction with n bidders having independent private values $v_i \sim F$ (where F is a continuous CDF on $[0, \bar{v}]$), show that bidding $b_i = v_i$ (truthful bidding) is a dominant strategy, hence a BNE. Compare with the first-price auction BNE from Example **??**.

Exercise 10.6 (Verifying a Potential Function)**.** Consider the two-player game with $A_1 = A_2 = \{X, Y\}$ and payoffs:

	X	Y
X	$3, 2$	$1, 1$
Y	$2, 1$	$4, 3$

(a) Show this is an exact potential game by constructing the potential function Φ.

(b) Find all pure Nash equilibria using Φ.

(c) Verify that best-response dynamics converge from any initial profile.

Exercise 10.7 (MWU Convergence Rate)**.** Consider the zero-sum game Rock-Paper-Scissors:

	R	P	S
R	0	-1	1
P	1	0	-1
S	-1	1	0

(a) What is the unique NE and the value of the game?

(b) If both players use MWU with $\eta = \sqrt{\ln 3/T}$, bound the duality gap $\max_{a_1} g(a_1, \bar{\alpha}_2^T) - \min_{a_2} g(\bar{\alpha}_1^T, a_2)$ after $T = 1000$ rounds.

(c) How many rounds are needed for the duality gap to be below 0.01?

Exercise 10.8 (Folk Theorem Application)**.** Consider the two-player stage game:

	C	D
C	$5, 5$	$0, 7$
D	$7, 0$	$2, 2$

(a) Compute the feasible payoff set V and the minimax values.

(b) Apply the Nash folk theorem to determine which payoff vectors are achievable as NE payoffs for large δ.

(c) For the payoff $(5, 5)$, find the minimum discount factor under grim trigger.

(d) For the payoff $(4, 4)$, describe a strategy profile achieving it as an SPE and find the minimum δ.

Exercise 10.9 (Congestion Game with Three Players)**.** Three players share two resources $\{A, B\}$. Each player chooses exactly one resource. Delay functions: $d_A(k) = k$ and $d_B(k) = 2k - 1$.

(a) Write the Rosenthal potential function Φ.

(b) Enumerate all action profiles and compute Φ for each.

(c) Find all pure NE by maximizing Φ (equivalently, minimizing $-\Phi$).

(d) Verify that your NE are indeed equilibria by checking no player wants to deviate.

Exercise 10.10 (Separating vs. Pooling Equilibria)**.** Consider a signaling game with two sender types $\theta \in \{G, B\}$ (good, bad), $\mathbb{P}(\theta = G) = 1/3$, two messages $m \in \{H, L\}$ (high, low effort), and two receiver actions $a \in \{A, R\}$ (accept, reject). Payoffs:

$$u_S(G, H, A) = 3, \quad u_S(G, H, R) = -1, \quad u_S(G, L, A) = 4, \quad u_S(G, L, R) = 0,$$
$$u_S(B, H, A) = 3, \quad u_S(B, H, R) = -2, \quad u_S(B, L, A) = 4, \quad u_S(B, L, R) = 1,$$
$$u_R(\theta, m, A) = \begin{cases} 1 & \text{if } \theta = G, \\ -1 & \text{if } \theta = B, \end{cases} \qquad u_R(\theta, m, R) = 0.$$

(a) Does a separating PBE exist where type G sends H and type B sends L?

(b) Does a pooling PBE exist where both types send L?

(c) Does a pooling PBE exist where both types send H?

11 Summary of Key Results

Result	**Statement**	**Ref.**
Unraveling Theorem	Unique stage-game NE $\Rightarrow$ unique SPE of G^T: play NE every period	Thm ??
Nash Folk Theorem	Any feasible, strictly individually rational payoff is a NE payoff for δ near 1	Thm ??
SPE Folk Theorem	Same as Nash folk theorem but for SPE (under full dimensionality condition)	Thm ??
Fictitious Play (2×2)	Empirical frequencies converge to NE in 2×2 zero-sum games	Thm ??
Robinson's Theorem	Fictitious play converges to NE in all finite two-player zero-sum games	Thm ??
MWU Regret Bound	External regret $\leq 2\sqrt{T \ln |A_i|}$; hence MWU is no-regret	Thm ??
No-Regret $\Rightarrow$ NE	If both players use no-regret, time-avg. play converges to NE in zero-sum	Thm ??
BNE Existence	Every finite Bayesian game has a BNE in mixed strategies	Thm ??
PBE in Signaling Games	Every finite signaling game has a PBE	Prop 7.1
Potential $\Rightarrow$ Pure NE	Every finite potential game has a pure Nash equilibrium	Thm 8.1
Finite Improvement	Best-response dynamics converge in finite steps in potential games	Thm 8.2
Rosenthal's Theorem	Every congestion game is an exact potential game	Thm 8.4

Game Class	**MARL Implication**
Repeated games (general-sum)	Folk theorems $\Rightarrow$ many equilibria; independent learners may converge to any of them; sensitive to initialization and hyperparameters
Zero-sum games	Minimax + no-regret $\Rightarrow$ convergent self-play (NFSP, PSRO, CFR)
Bayesian games	Opponent modeling as Bayesian inference over types; principled exploration
Potential games	Independent best-response / Q-learning converges; difference rewards make cooperative MARL a potential game
Congestion games	Routing, resource allocation: pure NE exist, local search works